

PF Lab 11 Tasks (6)

Name: Hassan Ali Khan

Section: BCS-1K

Roll No: 25K-0516

Task 5

```
#include<stdio.h>

#include<string.h>

void reverseCode(char s[], int len){

    if(len==0){

        return;

    }

    printf("%c",s[len-1]);

    reverseCode(s,len-1);

    return;

}

int main()

{

    char s[20];

    printf("Enter code word:");

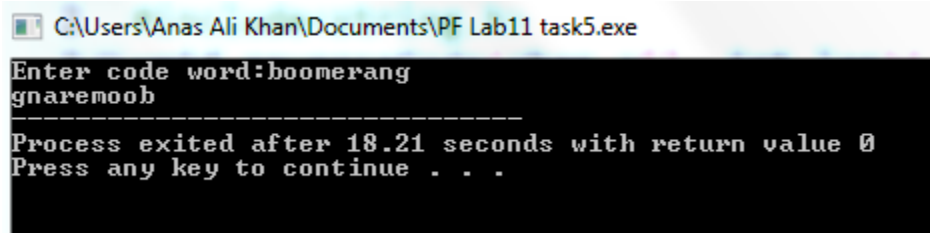
    gets(s);

    int len=strlen(s);

    reverseCode(s,len);

    return 0;

}
```



```
C:\Users\Anas Ali Khan\Documents\PF Lab11 task5.exe
Enter code word:boomerang
gnaremoob
-----
Process exited after 18.21 seconds with return value 0
Press any key to continue . . .
```

Task 6

```
#include<stdio.h>
```

```
int power(int num,int p){
```

```
    if(p==0){
```

```
        return 1;
```

```
    }
```

```
    int ans=num*power(num,p-1);
```

```
    return ans;
```

```
}
```

```
int main()
```

```
{
```

```
    int num,p;
```

```
    printf("Enter a number:");
```

```
    scanf("%d",&num);
```

```
    printf("\nEnter power(1-3):");
```

```
    scanf("%d",&p);
```

```
    printf("Answer=%d",power(num,p));
```

```
    return 0;
```

```
}
```

C:\Users\Anas Ali Khan\Documents\PF Lab11 task6.exe

Enter a number:4

Enter power(1-3):2

Answer=16

Process exited after 11.91 seconds with return value 0
Press any key to continue . . .

Task 7

```
#include<stdio.h>
```

```
struct university{
```

```
    char department[10];
```

```
    int block;
```

```
    char name[20];
```

```
    int id;
```

```
    float cgpa;
```

```
};
```

```
int main()
```

```
{
```

```
    struct university student;
```

```
    printf("Department:");
```

```
    scanf("%s",student.department);
```

```
    printf("\nBlock:");
```

```
    scanf("%d",&student.block);
```

```
    printf("\nName:");
```

```
    scanf("%s",student.name);
```

```
    printf("\nID:");
```

```
    scanf("%d",&student.id);
```

```
    printf("\nCGPA:");
```

```
    scanf("%f",&student.cgpa);
```

```
    printf("\n");
```

```

    printf("Department:%s\n",student.department);

    printf("Block:%d\n",student.block);

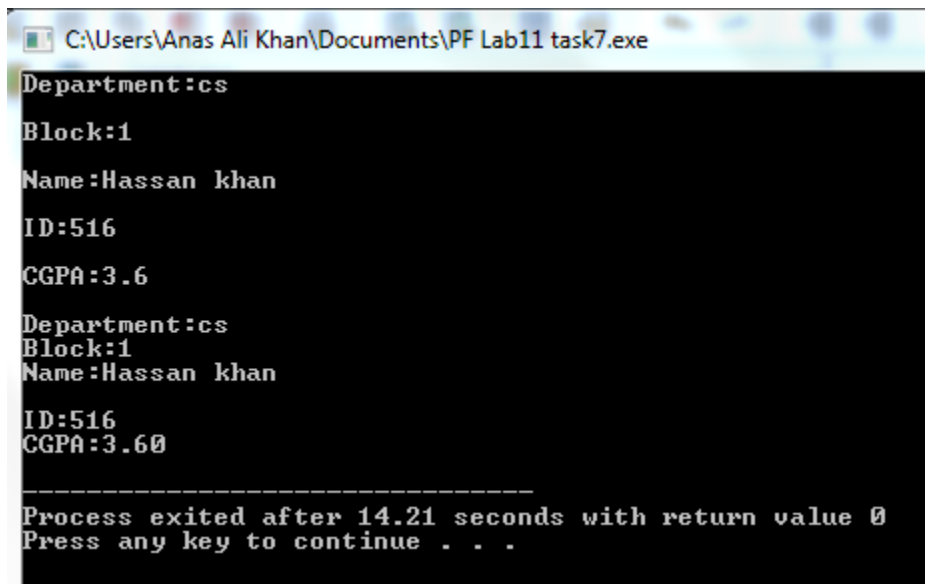
    printf("Name:%s\n",student.name);

    printf("ID:%d\n",student.id);

    printf("CGPA:%.2f\n",student.cgpa);

    return 0;
}

```



```

C:\Users\Anas Ali Khan\Documents\PF Lab11 task7.exe
Department:cs
Block:1
Name:Hassan khan
ID:516
CGPA:3.6
Department:cs
Block:1
Name:Hassan khan
ID:516
CGPA:3.60
-----
Process exited after 14.21 seconds with return value 0
Press any key to continue . . .

```

Task 8

```
#include<stdio.h>
```

```

struct employees{

    int id;

    char name[20];

    int salary;

    char position[10];

```

```
};
```

```
int main()
```

```
{
```

```
    struct employees emp[5];
```

```
    for(int i=0;i<5;i++){
```

```
        printf("Employee %d\n",i+1);
```

```
        printf("Id:");
```

```
        scanf("%d",&emp[i].id);
```

```
        getchar();
```

```
        printf("\nName:");
```

```
        fgets(emp[i].name,20,stdin);
```

```
        printf("\nSalary:");
```

```
        scanf("%d",&emp[i].salary);
```

```
        getchar();
```

```
        printf("\nPosition:");
```

```
        fgets(emp[i].position,10,stdin);
```

```
        printf("\n");
```

```
    }
```

```
    for(int i=0;i<5;i++){
```

```
        if(emp[i].salary>emp[i+1].salary){
```

```
            printf("Employee %s id %d of position %s has highest salary of  
%d",emp[i].name,emp[i].id,emp[i].position,emp[i].salary);
```

```
        }
```

```

    }

    return 0;
}

```

```

C:\Users\Anas Ali Khan\Documents\PF Lab11 task8.exe
Employee 1
Id:123
Name:Hassan ali
Salary:23000
Position:HR
Employee 2
Id:434
Name:Owais ali
Salary:20000
Position:Manager
Employee 3
Id:saad khan
Name:
Salary:22000
Position:manager
Employee 4
Id:111
Name:xyz
Salary:19000
Position:waiter
Employee 5
Id:333
Name:vvv
Salary:10000
Position:waiter
Employee Hassan ali
id 123 of position HR
has highest salary of 23000Employee aad khan
id 0 of position manager
has highest salary of 22000Employee xyz
id 111 of position waiter
has highest salary of 19000Employee vvv
id 333 of position waiter
has highest salary of 10000
-----
Process exited after 112.8 seconds with return value 0
Press any key to continue . . .

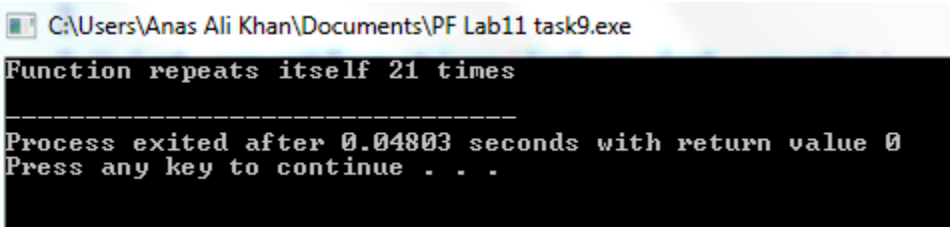
```

Task 9

```
#include<stdio.h>
```

```
int countFunction(int n,int count){  
    if(count==n){  
        return count;  
    }  
    countFunction(n,count+1);  
}
```

```
int main()  
{  
    int n=21, count=0;  
    printf("Function repeats itself %d times\n",countFunction(n,count));  
    return 0;  
}
```



```
C:\Users\Anas Ali Khan\Documents\PF Lab11 task9.exe  
Function repeats itself 21 times  
-----  
Process exited after 0.04803 seconds with return value 0  
Press any key to continue . . .
```

Task 10

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define MAX_MOVIES 10
```

```
#define MAX_LENGTH 50
```



```
struct Movie {  
  
    char title[MAX_LENGTH];  
  
    char genre[MAX_LENGTH];  
  
    char director[MAX_LENGTH];  
  
    int release_year;  
  
    float rating;  
  
};
```

```
struct Movie movies[MAX_MOVIES];  
  
int movie_count = 0;
```

```
void addMovie();  
  
void searchByGenre();  
  
void displayMovies();
```

```
int main() {  
  
    int choice;  
  
  
    printf("=== Movie Database System ===\n");  
  
  
    do {  
  
        printf("\nMenu:\n");  
  
        printf("1. Add Movie\n");  
  
        printf("2. Search by Genre\n");
```

```
printf("3. Display All Movies\n");

printf("4. Exit\n");

printf("Enter your choice: ");

scanf("%d", &choice);

getchar();


switch(choice) {

    case 1:

        addMovie();

        break;

    case 2:

        searchByGenre();

        break;

    case 3:

        displayMovies();

        break;

    case 4:

        printf("Goodbye!\n");

        break;

    default:

        printf("Invalid choice! Please try again.\n");

}

} while(choice != 4);


return 0;
```

```
}
```

```
void addMovie() {
```

```
    if(movie_count >= MAX_MOVIES) {
```

```
        printf("Database is full! Cannot add more movies.\n");
```

```
        return;
```

```
    }
```

```
    printf("\n--- Add New Movie ---\n");
```

```
    printf("Enter title: ");
```

```
    fgets(movies[movie_count].title, MAX_LENGTH, stdin);
```

```
    movies[movie_count].title[strcspn(movies[movie_count].title, "\n")] = 0; // Remove newline
```

```
    printf("Enter genre: ");
```

```
    fgets(movies[movie_count].genre, MAX_LENGTH, stdin);
```

```
    movies[movie_count].genre[strcspn(movies[movie_count].genre, "\n")] = 0;
```

```
    printf("Enter director: ");
```

```
    fgets(movies[movie_count].director, MAX_LENGTH, stdin);
```

```
    movies[movie_count].director[strcspn(movies[movie_count].director, "\n")] = 0;
```

```
    printf("Enter release year: ");
```

```
    scanf("%d", &movies[movie_count].release_year);
```

```

printf("Enter rating (0.0-10.0): ");

scanf("%f", &movies[movie_count].rating);

getchar();

movie_count++;

printf("Movie added successfully!\n");
}

void searchByGenre() {

    char search_genre[MAX_LENGTH];

    int found = 0;

    if(movie_count == 0) {

        printf("No movies in database!\n");

        return;

    }

    printf("\nEnter genre to search: ");

    fgets(search_genre, MAX_LENGTH, stdin);

    search_genre[strcspn(search_genre, "\n")] = 0;

    printf("\n--- Movies in '%s' genre ---\n", search_genre);

    for(int i = 0; i < movie_count; i++) {

        if(strcasecmp(movies[i].genre, search_genre) == 0) {

```

```

        printf("Title: %s\n", movies[i].title);

        printf("Director: %s\n", movies[i].director);

        printf("Release Year: %d\n", movies[i].release_year);

        printf("Rating: %.1f\n", movies[i].rating);

        printf("-----\n");

        found = 1;
    }
}

if(!found) {
    printf("No movies found in this genre.\n");
}
}

void displayMovies() {
    if(movie_count == 0) {
        printf("No movies in database!\n");
        return;
    }

    printf("\n--- All Movies (%d) ---\n", movie_count);

    for(int i = 0; i < movie_count; i++) {
        printf("Movie #%d:\n", i + 1);
    }
}

```

```

        printf(" Title: %s\n", movies[i].title);

        printf(" Genre: %s\n", movies[i].genre);

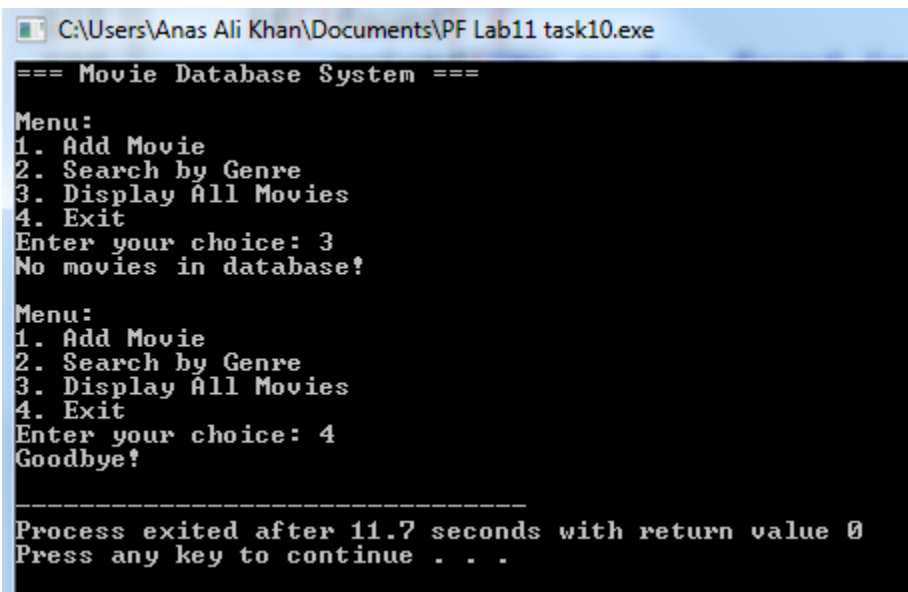
        printf(" Director: %s\n", movies[i].director);

        printf(" Release Year: %d\n", movies[i].release_year);

        printf(" Rating: %.1f/10.0\n", movies[i].rating);

        printf("-----\n");
    }
}

```



```

C:\Users\Anas Ali Khan\Documents\PF Lab11 task10.exe
=== Movie Database System ===
Menu:
1. Add Movie
2. Search by Genre
3. Display All Movies
4. Exit
Enter your choice: 3
No movies in database!

Menu:
1. Add Movie
2. Search by Genre
3. Display All Movies
4. Exit
Enter your choice: 4
Goodbye!

-----
Process exited after 11.7 seconds with return value 0
Press any key to continue . . .

```